

MLB Analysis: Insights with SQL

Introduction

This project uses SQL to analyze historical Major League Baseball (MLB) data and uncover trends in player development, team payrolls, career longevity, and player demographics. Through SQL queries involving joins, aggregations, window functions, and common table expressions (CTEs), the analysis identifies patterns across teams, schools, and decades, demonstrating how advanced SQL querying can extract meaningful insights from relational data. This project uses the following dataset: [MLB Data](#).

Key Insights & Takeaways

- The number of schools contributing players to MLB increased from **2 in the 1860s** to a peak of **494 in the 1990s**, reflecting a substantial expansion in the diversity of schools represented in the league. See [Figure 1](#) for query.
- The **top three schools** by total MLB player contributions are:
 - **University of Texas at Austin – 107 players**
 - **University of Southern California – 105 players**
 - **Arizona State University – 101 players**
 - See [Figure 2](#) for query.
- The **University of Texas at Austin** ranked among the **top three player-producing schools in six decades**, tying with **College of the Holy Cross** for the most top-three appearances. The **University of Southern**

California followed closely, appearing in the top three across **five decades**.

See [Figure 3](#) for query.

- The **San Francisco Giants**, **Los Angeles Angels**, and **New York Yankees** recorded the **highest average annual payrolls**, with an average annual spending of approximately **\$371 million** across the three teams. See [Figure 4](#) for query.
- The first teams to reach or exceed **\$1 billion in cumulative payroll** were the **New York Yankees (2003)**, **Boston Red Sox (2004)**, and **Atlanta Braves (2005)**. See [Figure 5](#) for query.
- Average MLB career length was **4.3 years** for players debuting in the **1870s** and fluctuated consistently to **1.2 years** for players debuting in the **2010s**, representing a decrease of **3.1 years**. See [Figure 6](#) for query.
- This decline in career length coincides with an increase in the **average debut age** (22 in 1870 and 24 in 2010) and a decrease in the **average retirement age** (28 in 1870 and 25 in 2010), resulting in progressively shorter playing careers. See [Figure 7](#) for query.
- The **longest career spent with a single team** was **23 years**, achieved by **two players**. See [Figure 8](#) for query.
- The **San Francisco Giants** had the **highest percentage of switch hitters**, with **18.5%** of players batting from both sides of the plate. See [Figure 9](#) for query.
- Average **debut weight** has increased consistently since the **1870s**, with the largest decade-over-decade increase occurring in the **2000s**, when average debut weight was **12 pounds higher** than in the 1990s. Average **debut height** has also increased over time, although the changes have been much more gradual. See [Figure 10](#) for query.

Queries

Figure 1. Number of player contributions by decade

```
10 SELECT FLOOR(yearID / 10) * 10 AS decade,
11         COUNT(DISTINCT schoolID) AS num_of_schools
12 FROM     schools
13 GROUP BY 1
14 ORDER BY 1;
15
```

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Result Grid Filter Rows: Search Export:

	decade	num_of_schools
	1860	2
	1870	14
	1880	34
	1890	89
	1900	148
	1910	178
	1920	196
	1930	162
	1940	142
	1950	176
	1960	301
	1970	427
	1980	473
	1990	494
	2000	372
	2010	57

Figure 2. Top 3 schools in player contribution

```
16 SELECT name_full AS school_name,
17        COUNT(DISTINCT playerID) AS num_of_players
18 FROM schools s INNER JOIN school_details sd
19        ON s.schoolID = sd.schoolID
20 GROUP BY 1
21 ORDER BY 2 DESC;
```

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Result Grid Filter Rows: Search Export: Fetch rows

school_name	num_of_players
University of Texas at Austin	107
University of Southern California	105
Arizona State University	101

Figure 3. Top 3 schools with most decades of player contributions

```
23 WITH psd AS (SELECT FLOOR(yearID / 10) * 10 AS decade,
24                     name_full AS school_name,
25                     COUNT(DISTINCT playerID) AS num_of_players,
26                     DENSE_RANK()
27                     OVER(
28                         PARTITION BY FLOOR(yearID / 10) * 10
29                         ORDER BY COUNT(DISTINCT playerID) DESC) AS top_schools
30 FROM schools s
31 INNER JOIN school_details sd
32     ON s.schoolID = sd.schoolID
33 GROUP BY 1,2
34 ORDER BY 1 DESC)
35
36 SELECT school_name,
37        SUM(num_of_players) total_players,
38        COUNT(decade) num_of_decades
39 FROM psd
40 WHERE top_schools IN (1,2,3)
41 GROUP BY school_name
42 ORDER BY 3 DESC;
```

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Result Grid Filter Rows: Search Export:

school_name	total_players	num_of_decades
University of Texas at Austin	61	6
College of the Holy Cross	67	6
University of Southern California	85	5

Figure 4. Top 3 teams in annual spending

```
49 WITH t1 AS (SELECT yearID, teamID, SUM(salary) AS total_spending
50 FROM salaries
51 GROUP BY 1,2
52 ORDER BY 1,2),
53
54 t2 AS (SELECT teamID,
55 ROUND(
56 AVG(total_spending) / 1000000, 2
57 ) AS avg_spend_millions
58 FROM t1
59 GROUP BY 1
60 ORDER BY 2 DESC)
61
62 SELECT teamID, avg_spend_millions
63 FROM (
64 SELECT teamID, avg_spend_millions,
65 NTILE(5) OVER() AS spend_percentile
66 FROM t2) sp
67 WHERE spend_percentile = 1;
68
```

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Result Grid Filter Rows: Search Export:

teamID	avg_spend_millio...
SFG	143.51
LAA	118.47
NYA	109.44

Figure 5. First 3 teams to reach \$1 billion in cumulative spending

```
82 WITH t1 AS (SELECT teamID,
83                    yearID,
84                    ROUND(
85                        SUM(salary) / 1000000000, 2)
86                    AS total_yearly_spend_bil
87                FROM salaries
88                GROUP BY 1,2),
89
90 t2 AS (SELECT teamID,
91              yearID,
92              SUM(total_yearly_spend_bil)
93              OVER(PARTITION BY teamID ORDER BY yearID
94                  ) AS cumulative_yearly_spend_bil
95              FROM t1)
96
97 SELECT teamID team,
98        MIN(yearID) AS year,
99        MIN(cumulative_yearly_spend_bil) cumulative_sum_billion
100 FROM t2
101 WHERE cumulative_yearly_spend_bil >= 1
102 GROUP BY 1
103 ORDER BY 2;
```

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Result Grid Filter Rows: Search Export:

team	year	cumulative_sum_billion
NYA	2003	1.07
BOS	2004	1.00
LAN	2005	1.07
ATL	2005	1.05

Figure 6. Average career length by debut decade

```
113 WITH bd AS (SELECT *,
114                CAST(CONCAT(birthYear,'-',birthMonth,'-',birthDay) AS DATE) AS birth_date
115                FROM players)
116
117
118 SELECT debut_decade,
119        AVG(career_length_years) avg_career_len
120 FROM
121 (
122     SELECT playerID,
123            nameGiven,
124            FLOOR(YEAR(debut) / 10) * 10 AS debut_decade,
125            FLOOR(YEAR(finalGame) / 10) * 10 AS finalGame_decade,
126            TIMESTAMPDIFF(YEAR, birth_date, debut) AS starting_age,
127            TIMESTAMPDIFF(YEAR, birth_date, finalGame) AS ending_age,
128            TIMESTAMPDIFF(YEAR, debut, finalGame) AS career_length_years
129     FROM bd ) t1
130 GROUP BY 1
131 ORDER BY 1;
```

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Result Grid Filter Rows: Search Export:

debut_deca...	avg_career_len
NULL	NULL
1870	4.2790
1880	3.4212
1890	3.6236
1900	3.6190
1910	3.5285
1920	3.9055
1930	5.0366
1940	4.2817
1950	5.0897

Figure 7. Average starting and ending age by decade

```
WITH bd AS (SELECT *,
                  CAST(CONCAT(birthYear,'-',birthMonth,'-',birthDay) AS DATE) AS birth_date
              FROM    players)

117
118 SELECT  debut_decade,
119          AVG(career_length_years) avg_career_len,
120          AVG(starting_age) AS avg_starting_age,
121          AVG(ending_age) AS Avg_ending_age
122 FROM
123 (
124     SELECT  playerID,
125             nameGiven,
126             FLOOR(YEAR(debut) / 10) * 10 AS debut_decade,
127             FLOOR(YEAR(finalGame) / 10) * 10 AS finalGame_decade,
128             TIMESTAMPDIFF(YEAR, birth_date, debut) AS starting_age,
129             TIMESTAMPDIFF(YEAR, birth_date, finalGame) AS ending_age,
130             TIMESTAMPDIFF(YEAR, debut, finalGame) AS career_length_years
131     FROM    bd ) t1
132 GROUP BY 1
133 ORDER BY 1;
```

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Result Grid Filter Rows: Search Export:

debut_deca...	avg_career_len	avg_starting_a...	Avg_ending_age
NULL	NULL	NULL	NULL
1870	4.2790	22.5329	28.3599
1880	3.4212	22.6816	27.1211
1890	3.6236	23.3531	27.6290
1900	3.6190	24.0690	28.0149
1910	3.5285	23.1043	26.9255
1920	3.9055	24.0705	28.2629
1930	5.0366	24.0010	29.3410
1940	4.2817	24.6233	29.2320

Figure 8. *Players over a decade-long career with loyalty to one team*

```
196 • SELECT      p.playerID,
197                team_count,
198                TIMESTAMPDIFF(YEAR, debut, finalGame) AS career_length_years
199 FROM          players p
200 INNER JOIN (
201             SELECT playerID,
202                    COUNT(DISTINCT teamID) AS team_count
203             FROM    salaries
204             GROUP BY 1) tc
205 ON            p.playerID = tc.playerID
206 WHERE         team_count = 1
207              AND TIMESTAMPDIFF(YEAR, debut, finalGame) > 10
208 ORDER BY     3 DESC;
209
```

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Result Grid Filter Rows: Search Export:

playerID	team_count	career_length_ye...
fiskca01	1	23
rosepe01	1	23

Figure 9. Team with the highest percentage of switch hitters

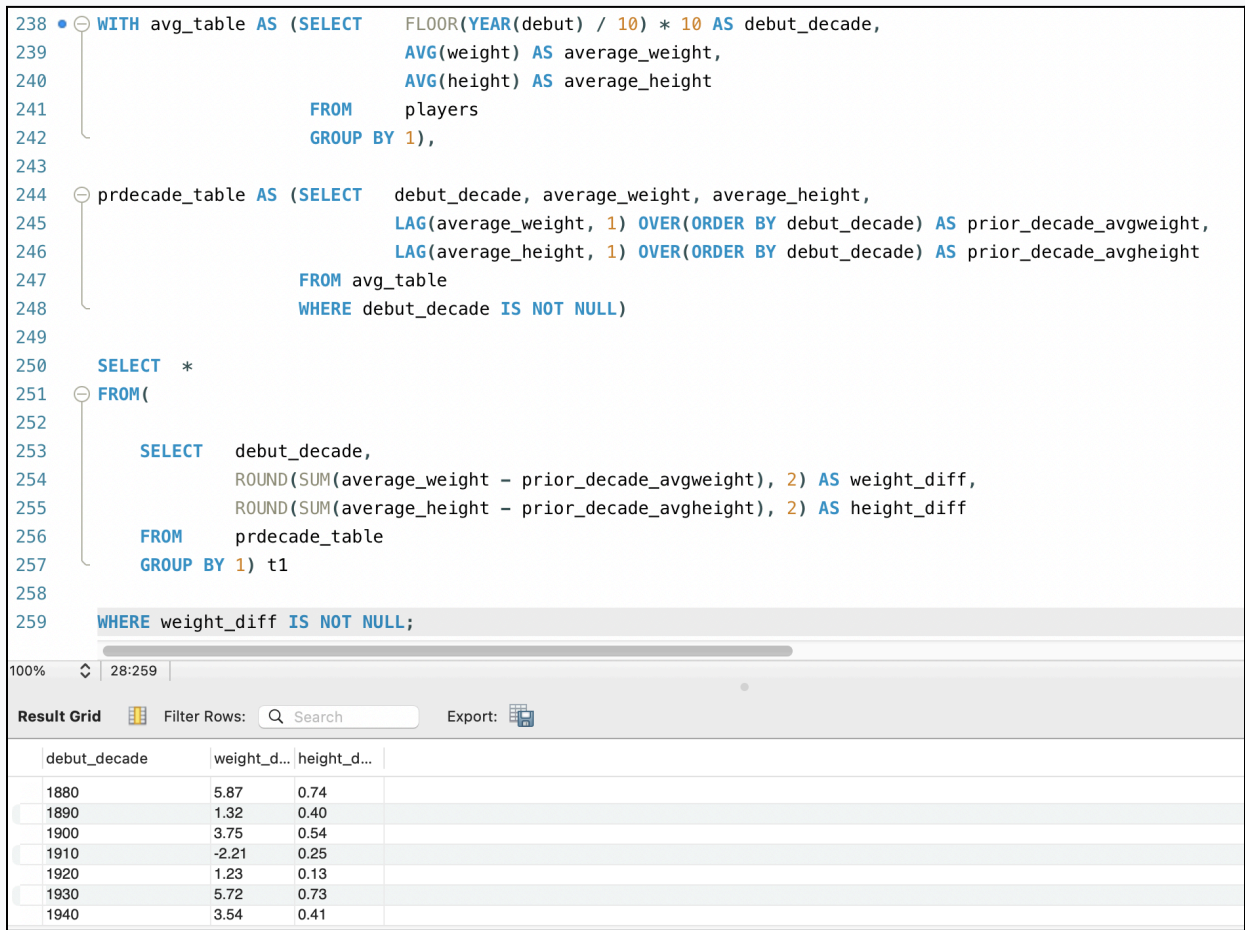
```
228 • SELECT s.teamID,  
229         ROUND(SUM(CASE WHEN bats = 'R' THEN 1 ELSE 0 END) / COUNT(s.playerID) * 100, 1) AS bats_right_percent,  
230         ROUND(SUM(CASE WHEN bats = 'L' THEN 1 ELSE 0 END) / COUNT(s.playerID) * 100, 1) AS bats_left_percent,  
231         ROUND(SUM(CASE WHEN bats = 'B' THEN 1 ELSE 0 END) / COUNT(s.playerID) * 100, 1) AS bats_both_percent  
232 FROM   salaries s INNER JOIN players p  
233       ON s.playerID = p.playerID  
234 GROUP BY 1  
235 ORDER BY 4 DESC;  
236
```

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Result Grid Filter Rows: Search Export:

teamID	bats_right_perc...	bats_left_perc...	bats_both_perc...
SFG	55.6	25.9	18.5
LAA	68.2	16.6	15.2

Figure 10. Decade-over-decade debut height and weight difference



debut_decade	weight_d...	height_d...
1950	2.00	0.41
1960	1.46	0.41
1970	0.18	0.19
1980	1.65	0.27
1990	6.19	0.15
2000	12.00	0.19
2010	1.43	-0.07